

Modified Syllabus w.e.f Oct 2021 onwards

CIVIL ENGINEERING AND APPLIED MECHANICS DEPARTMENT

B. E. I YEAR (4YDC)

CE: 10003: FUNDAMENTALS OF CIVIL ENGINEERING AND
APPLIED MECHANICS

CREDITS:

PERIOD PER WEEK			CREDITS			MAXIMUM MARKS				TOTAL MARKS
T	P	Tu	T	P	Tu	THEORY		PRACTICAL		
						CW	END SEM	SW	END SEM	
4	2	-	3	1	-	30	70	40	60	200

PRE-REQUISITE: XII standard (HSSC)

COURSE OBJECTIVES:

1. To study of all force systems, equilibrium of force system
2. To learn the concepts of centroids, centers of gravity and moments of inertia of simple and complex geometric shapes.
3. To learn the behavior of beams, frame and trusses under different loading conditions and to determine the shear force and bending moment diagram.
4. To study methods and equipments for chaining, linear & angular measurements.
5. To understand the fundamental concepts of levelling.

COURSE OUTCOMES:

1. To identify and analyze a system of forces, determine forces in members of trusses and calculate support reactions for beam subjected to various types of loading.
2. To determine the Center of Gravity and moment of Inertia of a given Plane Areas.
3. Determine and analyze the Shear force, Bending moment of beams and analyze the trusses and problems related to frictions
4. Understand the working principles of chaining instruments and measuring distance, perpendicular and oblique offsets from the given reference line
5. Apply the knowledge of surveying and leveling in different operations in civil engineering projects.

COURSE CONTENTS:

THEORY:

PART - A: (67 % Weightage)

APPLIED MECHANICS

UNIT-1

Forces and Equilibrium: Graphical and Analytical Treatment of Concurrent and Non-concurrent Co-planar forces, Free Body Diagram, Force Diagram and Bow's notations.
Application of Equilibrium Concepts: Frictional force in equilibrium problems.

UNIT-2

Centre of Gravity and Moment of Inertia: Centroid and Centre of Gravity, Moment of Inertia of Area and Mass, Radius of Gyration, Introduction to Product of Inertia and Principle Axes.
Power Transmission and its applications

UNIT-3

Beams: Types of Beams: Simply Supported Beam, Overhanging Beam, Cantilever Beam. Types of Supports of a Beam or Frame: Roller, Hinged and Fixed Supports. Load on the Beam or Frame: Different Types of Loading. Support Reaction of a Beam or Frame: Analytical Method. Analysis of Plane Trusses: Method of Joints, Method of Sections. Support Reactions
Introductions to shear force and Bending Moment and to draw shear force and bending moment diagram for UDL and Point Load for Cantilever and Simply supported beam without overhang.

PART - C: (33 % Weightage)

SURVEYING:

UNIT-4

Linear Measurements: Chain and Tape Surveying, Errors and Correction, Obstacles, Area Measurement by Planimeter.

Angular Measurements: Bearing, Prismatic Compass, Local Attraction, Declination, Bowditch rule of correction for traverse, Accuracy and Precision.

UNIT-5

Levelling: Types of Levels, Levelling Methods, Height of Instrument and Rise and Fall Method, Measurements, Recording, Reciprocal Levelling, Contours and Properties.

ASSESSMENT:

Continuous: Two midterm tests in a semester and a makeup test if required, Evaluation of Practical calculations & drawing sheets, internal submission and Viva Voice examination by internal examiner during, Semester-end: Theory examination of 3 Hours duration and Practical Viva Voice Examination by external examiner

PRACTICALS:

LAB COURSE OBJECTIVES:

1. To acquire practical knowledge on handling basic survey equipment's
2. To possess knowledge about chain, compass surveying and open traverse survey.
3. To have the ability to understand leveling and area of irregular figures
4. To study the principles of friction for different surfaces on horizontal plane and inclined plane.
5. To study and analyze the polygon law of forces.

LAB COURSE OUTCOMES:-

1. Understand the working principles of chaining instruments and measuring distance, perpendicular and oblique offsets from the given reference line.
2. Understand the principle of prismatic compass, open traverse survey.
3. Compute the RL of different points on the ground by fly and differential leveling and also compute the area of irregular figure.
4. Apply the principles of friction for different surfaces on horizontal plane and inclined plane.
5. Apply the analytical method to verify and compare the polygon law of forces.

List of Experiments:

1. To study the colour/sign conventions and various instruments used in chaining.
2. To determine the horizontal distance between two points by doing ranging and chaining.
3. To measure perpendicular and oblique offsets from a given reference line.
4. To study prismatic compass and perform open traverse survey.
5. To do traversing of any regular figure with the help of compass and eliminate local attraction.
6. To determine RL of different points on the ground by fly or differential leveling and compute by height of instrument (HI) method and rise and fall method
7. To determine area of an irregular figure with the help of a planimeter
8. To verify the principal of moment using bell crank level
9. To determine the coefficient of friction for different surface on horizontal plane.
10. To determine the coefficient of friction for different surface on inclined plane
11. Verify the polygon law of forces and compare with analytical method.

ASSESSMENT:

Sessional work will be based on syllabus of Engineering Mechanics and Surveying.

TEXT BOOKS RECOMMENDED:

1. Rajput R.K., *Engineering Mechanics*, Dhanpat Rai and Sons New Delhi.
2. D.S. Bedi, *Engineering Mechanics*, Khanna Book Publishing Co. (P) Ltd., Delhi
3. R. S. Khurmi, *Engineering Mechanics*, S.Chand Publishing
4. R.K. Bansal, *A Textbook of Engineering Mechanics*, Laxmi Publications
5. *Engineering Mechanics*, Sharma, Pearson.
6. Prasad I. B., *Applied Mechanics*, Khanna Publication New Delhi
7. Rammamurtham S., *Applied Mechanics*, Dhanpat Rai and Sons New Delhi
8. Duggal S.K., *Surveying Vol. 1*, Tata McGraw- Hill Education New Delhi
9. Punmia, B.C., *Surveying*, Laxmi Publications, New Delhi.

REFERENCE BOOKS:

1. Kumar KL, *Engineering Mechanics*, Tata McGraw- Hill Education New Delhi, Edition 4th, 2011
2. Ferdinand.P. Beer, E. Russell, Jr Johnston., David Mazurek, Philip J Cornwell, "*Vector Mechanics for Engineers: Statics and Dynamics*", McGraw – Hill Education New Delhi Edition 11th 2005.
3. Timoshenko, and Young D.H., "*Engineering Mechanics*", Tata Mc-Graw Hill Education New Delhi Edition 4th 2007.
4. Chanchandramouli P.N., *Engineering Mechanics*, PHI Learning Private Limited New Delhi Edition 2011

$$\frac{60}{\sin \theta + 0.577 \cos \theta} = \frac{100}{1.732 \cos \theta - \sin \theta}$$

or $1.732 \cos \theta - \sin \theta = 1.67 \sin \theta + 0.96 \cos \theta$

or $0.772 \cos \theta = 2.67 \sin \theta$

or $\tan \theta = \frac{0.772}{2.67} = 0.289$

or $\theta = \tan^{-1}(0.289) = 16.1^\circ$

$\therefore$ Tension induced in the string,

$$T = \frac{60}{\sin 16.1^\circ + 0.577 \cos 16.1^\circ} = 72.14 \text{ N (Ans.)}$$

HIGHLIGHT

1. **Parallelogram law of forces** : It states that if two forces acting simultaneously on a particle be represented in magnitude and direction by two adjacent sides of a parallelogram, their resultant may be represented in magnitude and direction by the diagonal of the parallelogram which passes their point of intersection.

$$\text{Analytically, } R = \sqrt{P_1^2 + P_2^2 + 2P_1 P_2 \cos \theta}$$

$$\text{and } \tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

2. **Triangle law of forces** : It states if two forces acting simultaneously on a body are represented in magnitude and direction by the two sides of a triangle taken in order, then their resultant may be represented in magnitude and direction by the third side taken in opposite order.
3. **Polygon law of forces** : It states that if a number of coplanar concurrent forces acting simultaneously on a body are represented in magnitude and direction by the sides of a polygon taken in order, then the resultant may be represented in magnitude and direction by the closing side of the polygon, taken in opposite order.
4. **Lami's theorem** states that if three coplanar forces acting on a point in a body keep it in equilibrium, then each force is proportional to the sine of the angle between the other two forces.

OBJECTIVE TYPE QUESTIONS

1. Fill in the blanks :

- (i) is something which changes or tends to change the state of rest or of uniform motion of a body in a straight line.
 - (ii) Magnitude, direction, sense and are the characteristics of a force.
 - (iii) The resistance to deformation, or change of shape, exerted by the material of body is called an force.
 - (iv) A force which prevents the motion, deformation of body is called a forces.
 - (v) An force is one which causes a body to move or change its shape.
 - (vi) is a method of designating a force by writing two capital letters one on either side of the force.
 - (vii) Forces whose lines of action pass through a common point are called forces.
 - (viii) A is a single force which can replace two or more forces and produce the same effect on the body as the forces.
 - (ix) A body whose dimensions are practically negligible is called a
 - (x) units are used by engineers for all practical purposes.
2. Say 'Yes' or 'No' :
- (i) A force system is a collection of forces acting on a body in one or more planes.
 - (ii) Coplanar concurrent collinear force system includes those forces whose vectors do not lie along the same straight line.
 - (iii) The force whose point of application is so small that it may be considered as a point is called a concentrated force.
 - (iv) A distributed force is one whose place of application is a point.
 - (v) The method of determination of the resultant of some forces acting simultaneously on a particle is called composition of forces.

Answers

1. (i) Forces; (ii) point of application; (iii) internal; (iv) passive; (v) active; (vi) Bow's notation; (vii) concurrent; (viii) resultant force; (ix) particle; (x) Gravitational.

2. (i) Yes, (ii) No, (iii) Yes, (iv) No, (v) Yes.

THEORETICAL QUESTIONS

- (a) Define a force. State the effects which a force may produce when it acts on the body.
(b) Enumerate characteristics of a force.
- (a) How are the forces classified?
(b) Define a 'force system'. Name the different force systems.
- Write a short note on 'Free body diagrams'.
- State the principle of 'Transmissibility of a force'.
- Explain briefly the following :
(i) Resultant force
(ii) Component of a force.
(iii) Principle of resolved parts.
- State the following laws of forces :
(i) Parallelogram law of forces ;
(ii) Triangle law of forces ;
(iii) Polygon law of forces.
- Discuss graphical and analytical methods for finding resultant of several coplanar concurrent forces.
- State equilibrium conditions for coplanar concurrent forces.
- State and prove Lami's theorem.

UNSOLVED EXAMPLES

- Find the components of a force of 100 N in directions, inclined to it at 30° and 40° on opposite sides. [Ans. 68.41 N; 53.2 N]
- Two forces equal to $2P$ and P respectively act on a particle; if the first be doubled, and the second increased by 12 N the direction of the resultant is unaltered, find the value of P . [Ans. 12 N]
- The angle between the two forces of magnitude 40 N and 30 N is 60° ; the 40 N force being horizontal. Determine the resultant in magnitude and direction; if (i) the forces are pulls and (ii) the 30 N force is push and 40 N

force is a pull. [Ans. (i) 60.8 N, $25^\circ 16'$; (ii) 36.05 N 46° with 40 N force)]

4. The resultant of two forces P and 15 N is 20 N inclined at 60° to the 15 N force. Find the magnitude and direction of P .

[Ans. $P = 18.03$ N, $106^\circ 52'$]

5. The following forces act at a point

- 50 N due North;
- 20 N North East;
- 30 N due East;
- 40 N in the direction inclined 30° East of South;
- 60 N in a direction inclined 60° South of West.

Find the resultant force. What angle does it make with East?

[Ans. 40.87 N at $33^\circ 20'$ S of E]

6. A particle is acted upon by the following forces :

- 20 N inclined 30° North of East
- 25 N towards North;
- 30 N towards North West;
- 35 N inclined 40° South of West.

Find magnitude and direction of the resultant force. [Ans. 45.59 N, $132^\circ 20'$ with East]

7. A particle is acted upon by three forces, equal to 5 N, 10 N and 13 N along three sides of an equilateral triangle, taken in order. Find graphically or otherwise the magnitude and direction of the resultant force.

[Ans. 7 N $20^\circ 47'$]

8. Forces of 2, 3, 5, $\sqrt{3}$ and 2 N respectively act at one of the angular points of regular hexagon towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force. [Ans. 11.11 N, $56^\circ 42'$]

9. Forces of 7, 1, 1 and 3 N respectively act at one of the angular points of regular pentagon towards the four other points, taken in order. Find the magnitude and direction of their resultant force. [Ans. 8.368 N, $31^\circ 55'$ with 7 N force]

10. A weight of 40 N is supported by two strings inclined 30° and 45° respectively to the horizontal. Find graphically or otherwise the tension in each cord.

[Ans. 29.30 N, 35.8 N]

11. A machine weighing 3 kN is supported by two chains attached to some point on the machine. One of these chains goes to an eye bolt in the wall and is inclined 30° to horizontal and the other goes to the hook in ceiling and is inclined at 45° to the horizontal. Find tensions in the two chains. [Ans. 2.196 kN, 2.690 kN]
12. A small ring is situated at the centre of a hexagon, and is supported by six strings drawn tight, all in the same plane and radiating from the centre of the ring, and each fastened to a different angular point of the hexagon. The tensions in four consecutive strings are 1 N, 3.5 N, 4.5 N and 3 N respectively. Find the tensions in the two remaining strings. [Ans. 1.5 N, 6.5 N]
13. A body of weight 4.5 N is suspended by two strings 18 cm and 61 cm long and other ends being fastened to the extremities of rod of length 63.6 cm. If the rod be so held that the body hangs immediately below its middle point, find out the tensions of the strings. [Ans. 4.32 N, 1.26 N]
14. In a simple jib crane, the vertical post is 2.4 m high, the jib is 3.9 metres long and the tie 2.1 metres long. Find the forces on the jib and rod when a weight of 2500 N is supported at the crane end. [Ans. Jib = 4050 N (comp.); Tie = 2900 N (tension)]
15. A rope AB, 2.7 m long is connected at A and B to two points on the same level, 2.4 m apart. A load of 136 N is suspended from a point C on the rope 90 cm from A. What load connected to a point D on the rope, 60 cm from B, will be necessary to keep the portion CD level? [Ans. 272 N]
16. A sphere of radius 20 cm and weight 1 N is suspended against a smooth wall by a string of length 20 cm. The string joins a point in the wall and a point on the surface of the sphere. Find the inclination and the tension in the string and the reaction of the wall. [Ans. 60° , $T = \frac{2}{\sqrt{3}}$ N; $R = \frac{1}{\sqrt{3}}$ N]
17. A body of weight 6.5 N is suspended from the horizontal beam by strings, whose lengths are respectively 40 cm and 157.5 cm, the strings being fastened to the beam at two points 162.5 cm apart. What are the tensions in the strings? [Ans. 63 N and 16 N]
18. Two weights P and Q are fixed to a string ABCD at the points B and C , the ends A and D of which are fixed. Portions AB and CD make angles 50° and 20° respectively with the horizontal and portion BC remains horizontal. If Q is 12 N, find the value of P and tension in portion BC of the string. [Ans. 39.29 N; 32.97 N]
19. Two equal heavy spheres of 5 cm radius are in equilibrium with a smooth cup of 15 cm radius. Show that reaction between the cup and one sphere is double than that between the two spheres.
20. Solve for the reactions at L , M , N and S on the two cylinders shown in Fig. 2.104, assuming all surfaces to be smooth. [Ans. $R_L = 62$ N; $R_M = 200$ N; $R_N = 79.5$ N; $R_S = 62$ N]

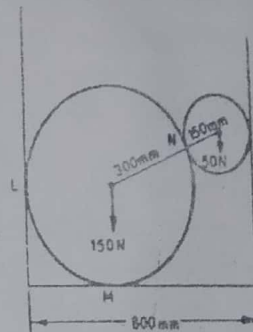


Fig. 2.104

21. A force P (Fig. 2.105) is applied at O to the string LOM . If the tension in each part of the string is 100 N, find the magnitude and direction of the force P for equilibrium conditions.

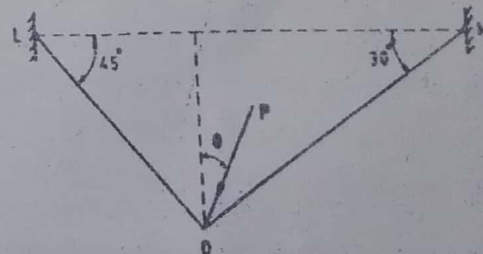


Fig. 2.105

[Ans. 122 N, $\theta = 7.5^\circ$]

22. A bar LM of weight 200 N is hinged at L and is pulled by a cable attached at M by a force P . Find the force P and the magnitude and direction of the reaction at L if the bar is in equilibrium position as shown in Fig. 2.106.

[Ans. 141.44 N , $R_L = 273.2\text{ N}$, 60°]

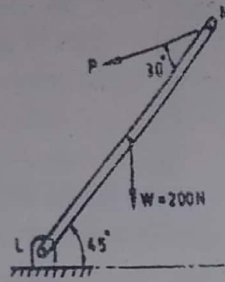


Fig. 2.106

23. A sphere of mass M rests in a V-groove (Fig. 2.107) whose sides are inclined at angle α and β to the horizontal. Another identical sphere of same mass M rests on the first sphere and in contact with the side inclined at angle α . Find the reaction R_M on the lower sphere at the point M .

[Ans. $R_M = \frac{2Mg \sin \alpha}{\sin (\alpha + \beta)}$]

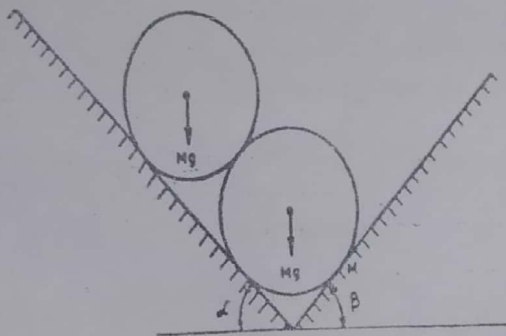


Fig. 2.107

24. Two steel cylinders of equal diameter are supported in a rectangular vessel (Fig. 2.108). Determine the reactions R_1 , R_2 and R_3 . Weight of each cylinder may be taken as 100 N .

[Ans. 200 , 150 , 112 N]

25. Two steel cylinders are supported in a right-angled wedge support as shown in Fig. 2.109. The side OL makes an angle of 30° with the horizontal. The diameters of the cylinders are 250 mm and 500 mm ; their weights being 100 and 400 N respectively. Determine the reaction R between the smaller cylinder and the side OL .

[Ans. 157 N]

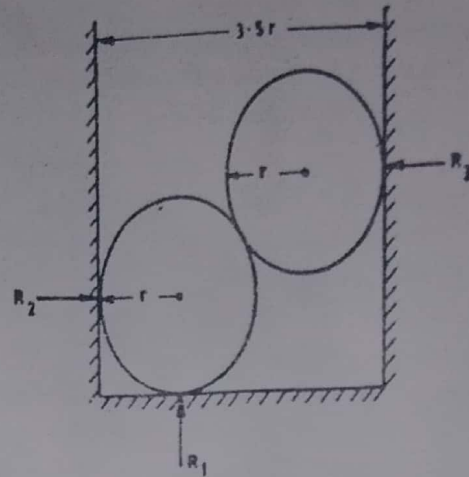


Fig. 2.108

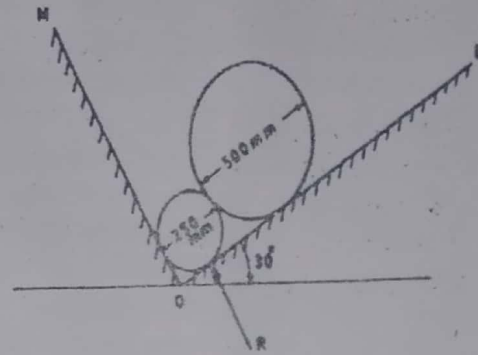


Fig. 2.109

26. Three spheres each weighing 100 N are arranged in a rectangular vessel of 500 mm base. If the diameter of each ball is 220 mm , determine:

(i) R_1 and R_2 ;

(ii) R ; (iii) R_n

Refer Fig. 2.110

[Ans. 149.8 N ; 41.2 N ; 64.8 N]

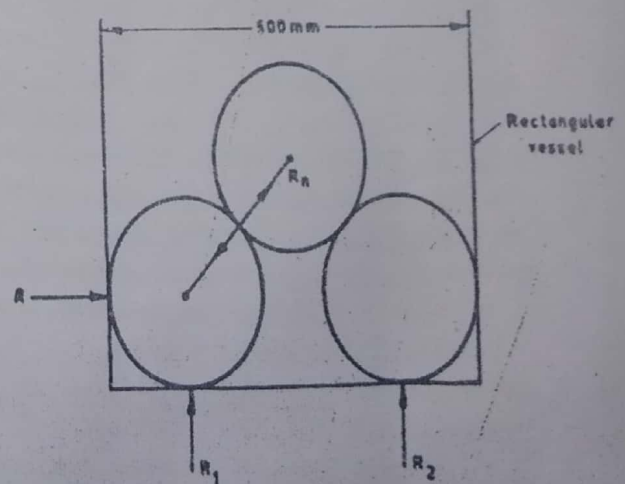


Fig. 2.110

Applying Lami's theorem, we get

$$\frac{R_B}{\sin(90^\circ + 2\theta)} = \frac{R_A}{\sin(180^\circ - \theta)} = \frac{10}{\sin(90^\circ - \theta)}$$

$$\text{or, } R_B = \frac{10}{\sin(90^\circ - \theta)} \times \sin(90^\circ + 2\theta)$$

$$\text{or, } R_B = \frac{10}{\cos \theta} \cos \theta \quad \dots(i)$$

$$\text{or, } R_A = \frac{10 \times \sin(180^\circ - \theta)}{\sin(90^\circ - \theta)} = \frac{10 \sin \theta}{\cos \theta}$$

$$\text{or, } R_A = 10 \tan \theta \quad \dots(ii)$$

$$\text{Also, } \Sigma M_A = 0$$

$$\text{or, } -R_B \times AB + W \times AG \cos \theta = 0 \quad \dots(iii)$$

Also,

$$AB = 2R \cos \theta = 2 \times 1 \times \cos \theta = 2 \cos \theta, \text{ and}$$

$$AG = \frac{3}{2} = 1.5 \text{ m}$$

Putting the values of R_B , AB and AG in (iii), we get

$$-\frac{10 \cos 2\theta}{\cos \theta} \times 2 \cos \theta + 10 \times 1.5 \cos \theta = 0$$

$$-\frac{10 \cos 2\theta}{\cos \theta} \times 2 \cos \theta + 15 \cos \theta = 0$$

$$-20 \cos 2\theta + 15 \cos \theta = 0$$

$$\text{or, } 20 \cos 2\theta - 15 \cos \theta = 0$$

$$\text{or, } \cos 2\theta - 0.75 \cos \theta = 0$$

$$\text{or, } 2 \cos^2 \theta - 1 - 0.75 \cos \theta = 0$$

$$\text{or, } 2 \cos^2 \theta - 0.75 \cos \theta - 1 = 0$$

$$\text{or, } \cos \theta = \frac{0.75 \pm \sqrt{0.75^2 + 4 \times 2 \times 1}}{4}$$

$$= \frac{0.75 \pm 2.926}{4} = 0.919 \text{ or } -0.544$$

$$\text{or, } \theta = 23.2^\circ \text{ or } 122.95^\circ$$

But $\theta = 122.95^\circ$ is not possible, therefore, correct value of $\theta = 23.2^\circ$ (Ans.)

HIGHLIGHTS

1. The rotational tendency of a force is called 'Moment'. The force multiplied by the perpendicular distance from the point to the line of action of the force is called *moment about that point* i.e. Moment of a force = force $\times$ perpendicular distance.
2. Conventionally, clockwise moments are taken as *positive* moments and *anti-clockwise* moments as negative moments.
3. Principle of moments states : "When a body acted upon by several forces is in rotational equilibrium, the sum of clock-wise moments of the forces about any point is equal to sum of the anti-clockwise moments of the forces about the same point."
4. The body can be in equilibrium if the algebraic sum of all the external forces and their moments about any point in their plane is *zero*. Mathematically, the conditions of equilibrium may be expressed as follows :
 - (i) $\Sigma P_x = 0$ (ΣP_x means sum of all the along forces X-axis (horizontal))
 - (ii) $\Sigma P_y = 0$ (ΣP_y means sum of all the forces along Y-axis (vertical))
 - (iii) $\Sigma M = 0$ (ΣM means sum of all the moments).
5. Varignon's theorem states : "The algebraic sum of the moments of two forces about any point in their plane is equal to moment of their resultant about that point."
6. Forces whose lines of action are parallel are called *parallel forces*. They are said to be like when they act in the *same sense*; they are said to be unlike when they act in *opposite sense*.
7. A couple is a pair of two equal and opposite forces acting on a body in such a way that the lines of action of the two forces are not in the same straight line.
The moment of couple is known as *torque* which is equal to one of the forces forming the couple multiplied by the arm of the couple.
8. Examples of couples in every day life :
 - (i) Opening or closing a water tap.
 - (ii) Turning the cap of a pen.

- (iii) Unscrewing the cap of an ink bottle.
 - (iv) Turning a screw driver.
 - (v) Steering a motor car.
 - (vi) Winding a watch or clock with a key.
9. Some important engineering applications of moments are :
- (i) The balance;
 - (ii) The steel yard;
 - (iii) Bell crank lever;
 - (iv) Lever safety valve; and
 - (v) Cranked lever.
6. "The algebraic sum of the moments of two forces about any point in their plane is equal to the moment of their resultant about that point." The above statement is called
- (a) Principle of moments
 - (b) Varignon's theorem
 - (c) Theorem of parallel axes
 - (d) Theorem of perpendicular axes.
7. A is a pair of two equal and opposite forces acting on a body in such a way that the lines of action of the two forces are not in the same straight line.

OBJECTIVE TYPE QUESTIONS

Choose the Correct Answer

1. The rotational tendency of a force is called
 - (a) shear force
 - (b) moment
 - (c) centroid
 - (d) couple.
2. Moment of a force is a quantity.
 - (a) scalar
 - (b) vector
 - (c) either (a) or (b)
 - (d) none of the above.
3. Moment of a force
 - (a) varies directly with its distance from the pivot
 - (b) varies inversely with its distance from the pivot
 - (c) is independent of its distance from the pivot
 - (d) none of the above.
4. "When a body acted upon by several forces is in rotational equilibrium, the sum of the clockwise moments of the forces about any point is equal to the sum of the anti-clockwise moments of the forces about the same point." The above statement is called
 - (a) Theorem of parallel axes
 - (b) Theorem of perpendicular axes
 - (c) Principle of moments
 - (d) None of the above.
5. The conditions of equilibrium for coplanar non-concurrent forces are
 - (a) $\sum H = 0, \sum V = 0$
 - (b) $\sum H = 0, \sum M = 0$
 - (c) $\sum V = 0, \sum M = 0$
 - (d) $\sum H = 0, \sum V = 0, \sum M = 0$
8. Which of the following is an example of a couple?
 - (a) Turning the cap of a pen
 - (b) Unscrewing the cap of an ink bottle
 - (c) Twisting a screw driver
 - (d) Steering a motor-car
 - (e) All of the above.
9. The principle of lever was developed by
 - (a) Archimedes
 - (b) Kelvin
 - (c) Watt
 - (d) Newton.
10. Which of the following is an example of lever?
 - (a) Crow bar
 - (b) A pair of scissors
 - (c) Fire tongs
 - (d) All of the above.
11. For true balance which of the following conditions must be satisfied ?
 - (a) The arms of the balance must be of equal size
 - (b) The weights of scale pans must be equal
 - (c) The fulcrum, the centre of rigid beam and c.g. of the beam including its connected parts must lie on the line perpendicular to the beam
 - (d) All of the above.

ANSWERS

1. (b); 2. (b); 3. (a); 4. (c); 5. (d); 6. (b);
7. (b); 8. (e); 9. (a); 10. (d); 11. (d).

THEORETICAL QUESTIONS

- Define the terms 'Moment' and Moment of a force.
 - State and explain the principle of moments.
- State equilibrium conditions for bodies under coplanar forces.
- State and prove Varignon's theorem.
- Explain the graphical method for finding the resultant of any number of like or unlike parallel forces.
- What is a couple?
 - Enumerate the properties of a couple.
- State some engineering applications of moments.
- Explain the method of finding resultant of a coplanar, non-concurrent, non-parallel force system.

UNSOLVED EXAMPLES

- Four forces 3, 4, 5 and 6 N respectively act in a clockwise direction along the sides of a square each side of which is 0.5 m. Find the value of the moments of these forces about (i) the centre of the square and (ii) the point of intersection of the forces 3 N and 6 N.

[Ans. 450 N.m; 450 N.m]

A man and a boy carry a weight of 250 N between them by means of a uniform pole 1.7 metres long and weighing 90 N. Where must the weight be placed so that man bear twice as much of the weight as that of the boy?

[Ans. 0.465 m]

- Forces act along the sides AB , AD , CB and CD of a square $ABCD$ equal to 5, 10, 15 and 20 N respectively. Also a force of $8\sqrt{2}$ N acts along BD . Find the algebraic sum of moments about the point A . The sides of the square being 2 metres long.
- A pair of nut crackers is 250 mm long and a nut is placed at a distance of 50 mm from the hinge. Find the force required to crack the nut if it requires a weight of 10 N for its cracking.

[Ans. 2 N]

- $ABCD$ is a square whose side is 2 m. Along AB , BC , CD and DA act forces equal to 1, 2, 8 and 5 N and along AC and DB forces equal to $5\sqrt{2}$ and $2\sqrt{2}$ N. Show that they are equivalent to a couple whose moment is equal to 16 Nm.

- Find out the effort required on one end of the iron crow bar of weight 100 N and length 5 metres to lift a weight of 2000 N on the other end. Fulcrum is kept at a distance of 4 metres from the end where effort is to be applied.

[Ans. 462.5 N]

- Find the equilibrium of the system of forces in magnitude and direction as shown in Fig. 3.161.

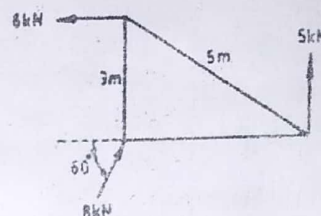


Fig. 3.161

[Ans. $R_e = 6.3$ kN acting clockwise from point A at an angle $17^\circ 48'$ with the horizontal]

- The lever of common steel yard without hook connections weighs 70 N and acts at 50 mm to the left of the fulcrum. The hook which carries the load, weighs 40 N and is also attached on the left hand side of the fulcrum at a distance of 120 mm. If the graduation mark 720 mm from the fulcrum reads 500 N, find

(i) The weight of the rider.

(ii) How far is the zero graduation from the fulcrum? [Ans. 94.8 N, 87.5 mm]

- The lever of a safety valve weighs 30 N and its c.g. lies at a distance of 80 mm from the centre of the valve which is at 70 mm from the fulcrum. If the diameter and weight of the valve are respectively 60 mm and 50 N, find the minimum weight that will be hung to the end of 750 mm long lever to conserve steam in the boiler at a pressure of 0.8 N/mm^2 .

[Ans. 205.8 N]

10. A lever safety valve 60 mm diameter is just on the point of blowing off steam when the weight of the valve is 10 N, weight on the lever is 300 N acting at 70 mm from the fulcrum and weight of the lever is 40 N and acting at 50 mm from the fulcrum towards the weight of 300 N. The valve is 40 mm from the fulcrum acting on the same side as weight. Find the intensity of pressure of the steam.

[Ans. $p = 0.208 \text{ N/mm}^2$]

11. A uniform ladder AB whose weight is 200 N and length 6 m rests against a smooth vertical wall making with it an angle of 30° . The other end rests on the ground surface. Find the reactions given to the ladder by the wall and the floor and their inclination to the vertical, when a man weighing 700 N climbs the ladder by a distance 1.5 m along the length of ladder.

[Ans. 332 N; 1536 N, $12^\circ 27'$]

12. Find the magnitude of two like parallel forces acting at a distance of 1 m apart which is equivalent to a force of 300 N acting at a distance of 200 mm from one of the forces.

[Ans. 240 N]

13. Find the magnitude and line of action of the resultant of parallel forces 7, 5 and 9 N in one direction and 12 N in the opposite direction acting at points A, B C and D respectively in a straight line where $AB = 120 \text{ mm}$, $BC = 360 \text{ mm}$ and $CD = 500 \text{ mm}$.

[Ans. Resultant = 9 N, acting at 760 mm from 7 N force]

14. Four coplanar forces equal to 20 N, 30 N, 50 N and 70 N are acting on a square body of each side equal to 1 m as shown in Fig. 3.162 Determine the magnitude, direction and position of the force which will keep the body in equilibrium.

[Ans. $R = 65.3 \text{ N}$ acting anti-clockwise from point C at an angle $20^\circ 30'$ with the horizontal and at a distance of 0.42 m]

15. The equilateral triangle shown in Fig. 3.163 has a side of 100 mm. Find

(i) The magnitude and direction of the resultant force.

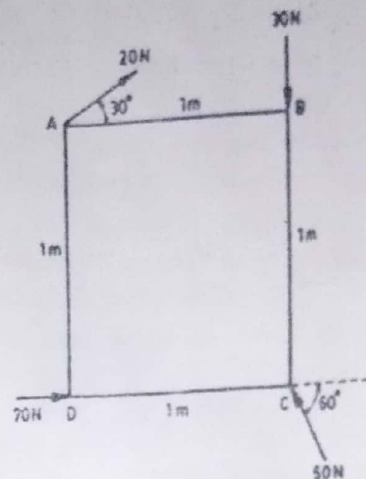


Fig. 3.162

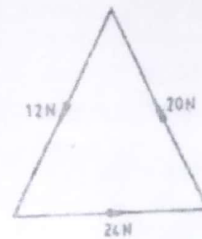


Fig. 3.163

(ii) The point at which the resultant cuts QR.

[Ans. 10.6 N, 41° to QR, 150 mm to right of R on QR produced]

16. The rectangle shown, Fig. 3.164, has $AB = 200 \text{ mm}$ and $BC = 150 \text{ mm}$. Calculate the magnitude, direction, and position of the resultant force.

[Ans. 10.2 kN, $11^\circ 19'$ to BA; cuts 40 mm from A on DA]

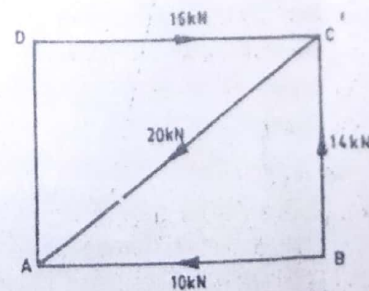


Fig. 3.164

17. $ABCD$ is a square each side being 500 mm and E is the mid point of AB . Forces of 3.18, 3.63, 5.44, 2.27, 4.10 and 2.72 N act on a body along the lines and directions of AB, EC, BC, BD, CA and DE respectively. Find the magnitude and direction and position with respect to $ABCD$ of the force required to keep the body in equilibrium.

[Ans. $R = 5.22$ N, acts in a direction making an angle of 197.25° with AD and at a distance $x = 274.5$ mm from E , so as to cause an anticlockwise couple]

18. A square $ABCD$ has forces acting along its sides as shown in Fig. 3.165. Find the values of P_1 and P_2 , if the system reduces to couple. Also find magnitude of the couple, if the sides of the square is 1 m.

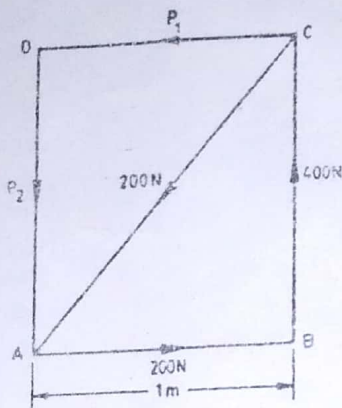


Fig. 3.165

[Ans. $P_1 = 58.6$ N, $P_2 = 258.6$ N; couple = 458.6 N-m (anticlockwise)]

19. $ABCD$ is a square of each side 200 mm and E is the middle point of AB . Forces of 21, 24, 36, 15, 27 and 18 N act on a body in the lines of directions AB, EC, BC, CA and DE respectively. Find the magnitude, direction and position of the resultant force.

[Ans. $R = 34.38$ N; $\theta = 17^\circ 3'$ (with horizontal); Position of R , $x = 113.3$ mm... moments about E]

20. A bar 4 m long and of negligible weight is acted upon by a vertical load of 600 N and a horizontal load of 300 N applied at positions as shown in Fig. 3.166. The ends of the bar are

in contact with a smooth vertical wall and a smooth incline. Determine the equilibrium position of the bar as defined by the angle θ it makes with the horizontal.

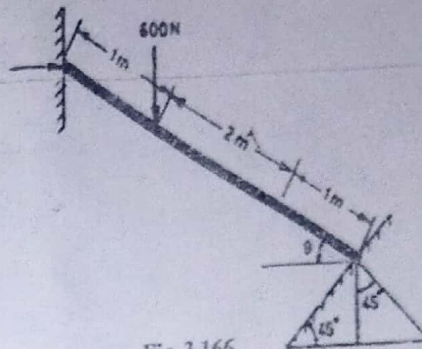


Fig. 3.166

[Ans. $\theta = 28.61^\circ$]

21. Bar LM of negligible weight is subjected to a vertical force of 300 N and a horizontal force of 150 N applied as shown in Fig. 3.167. Find the angle θ at which equilibrium exists. Assume smooth inclined surfaces. [Ans. 45°]

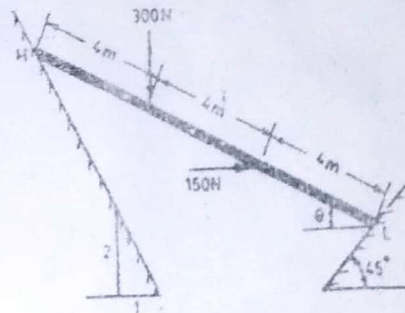


Fig. 3.167

22. A 12 m bar of negligible weight rests in a horizontal position on the smooth inclines as shown in Fig. 3.168. Compute the distance 'a' at which load $W = 100$ N should be placed from the point B to keep the bar horizontal.

[Ans. 4.83 m]

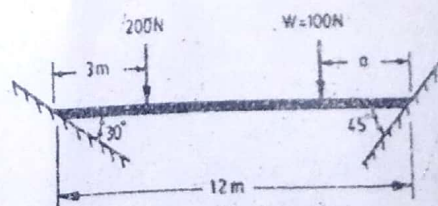


Fig. 3.168

23. A simply supported overhanging beam 15 metres long carries a system of loads as shown in Fig. 3.169. Determine the reactions at the supports.

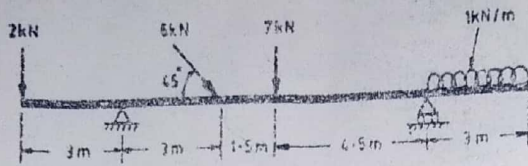


Fig. 3.169

[Ans. 7.75 kN; 9.27 kN acting at $26^\circ 36'$ with vertical]

24. Determine the reactions at the supports in case of the beam shown in Fig. 3.170.

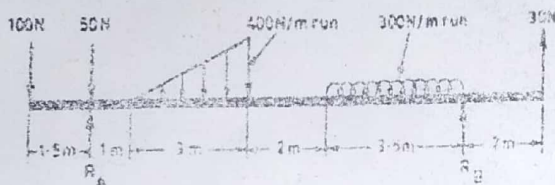


Fig. 3.170

[Ans. 776 N; 994 N]

25. Determine the reactions of the supports in case of the beam shown in Fig. 3.171.

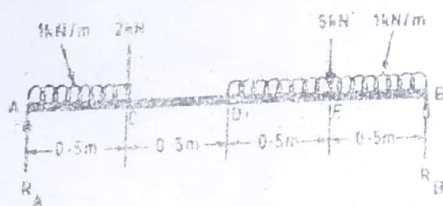


Fig. 3.171

[Ans. $R_A = 3.4$ kN; $R_B = 5.1$ kN]

26. The intensity of loading on a simply supported beam of 4 m span increases gradually from 30 kN/m run at one end to 130 kN/m run at the other as shown in Fig. 3.172. Determine the reactions at the supports.

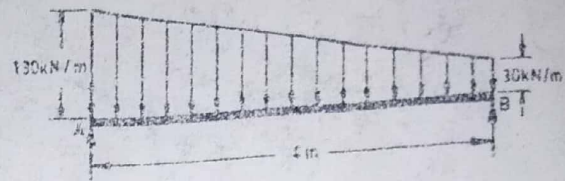


Fig. 3.172

[Ans. $R_A = 193.33$ kN and $R_B = 126.67$ kN]

27. Determine the reactions at the supports in case of the beam shown in Fig. 3.173.

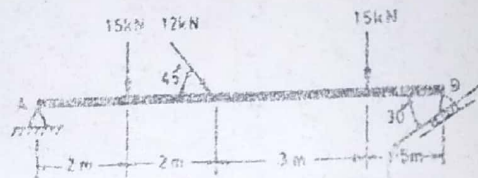


Fig. 3.173

[Ans. $R_A = 18.84$ kN, $\theta = 9^\circ 6'$ (with vertical); $R_B = 22.95$ kN]

28. Determine the reactions at the supports A and B of the beam loaded as shown in Fig. 3.174.

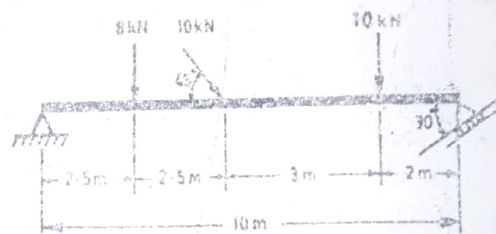


Fig. 3.174

[Ans. $R_B = 15.62$ kN; $R_A = 11.56$ kN and $\theta = 3.67^\circ$ (with vertical)]